Python Engineering - Class 01 Theory Guide

From Equation to Algorithm

This guide explains the programming, mathematical and/or engineering theory behind the exercise. The goal is to understand the model before memorizing syntax.

1. From equation to algorithm

A mathematical equation defines a relationship, but a program also needs inputs, validation, execution order and outputs.

Relationship / equation

```
q = 0.5 * rho * V^2
```

Connection to code

rho and speed are inputs; q is the output. The algorithm adds validation and display around the equation.

2. Physical meaning of dynamic pressure

Dynamic pressure represents the kinetic pressure scale associated with a moving fluid. It grows linearly with density and quadratically with speed.

Relationship / equation

```
q proportional to rho; q proportional to V^2
```

Connection to code

Doubling speed multiplies q by four, which is a useful manual sanity check.

3. Units and dimensional checking

Using SI units makes the result interpretable and helps catch mistakes.

Relationship / equation

```
(kg/m^3)*(m/s)^2 = kg/(m*s^2) = Pa
```

Connection to code

A result in Pa confirms that the implemented expression is dimensionally consistent.

4. Validation as part of engineering software

A formula can be mathematically executable with values that are physically meaningless. Software should separate mathematical possibility from physical validity.

Relationship / equation

```
rho >= 0; speed >= 0
```

Connection to code

The code checks inputs before evaluating the equation.

Manual validation method

For any exercise in this class, use the following protocol before trusting the output:

- 1. Write input values with units or types.
- 2. Predict sign, scale, type and shape of the result.
- 3. Manually calculate a small case when possible.
- 4. Run the code and compare against the prediction.
- 5. Change one input in a controlled way and explain the effect.

Connection to the rest of the course

This class establishes the equation-to-algorithm workflow used in every later numerical and aerospace problem.

Final summary

```
MODEL / CONCEPT
      |
      v
ALGORITHM
      |
      v
PYTHON IMPLEMENTATION
      |
      v
REFERENCE CHECK
      |
      v
ENGINEERING INTERPRETATION
```

> Syntax is the tool. The transferable skill is being able to model, structure, validate and interpret.